Gauge R&R Study Process Flow

Steps for conducting a complete MSA study

1. Plan Study
Define measurement, tolerance, study type (variable/attribute)

2. Select Parts
Select 10 parts spanning process range

3. Identify Appraisers
Select 2–3 operators who normally use gauge

4. Calibrate Gauge
Verify gauge is calibrated and in good condition

5. Randomize Order
Randomize measurement order to prevent bias

6. Conduct Measurements
Each operator measures each part 2–3 times (blind)

7. Analyze Data
Calculate EV, AV, GRR, %GRR, ndc

8. Interpret Results
Compare to acceptance criteria

9. Take Action
If unacceptable, identify root cause and improve

10. Document
Record results in quality system